

ENGINEERING CHEMISTRY (BAS102 / BAS202)

Unit-3 One Shot

Today's Target

- ✓ All topics of Unit-3

As per New
Syllabus
2022-2023

Electrochemistry and Batteries: Basic concepts of electrochemistry.

Batteries: Classification and applications of Primary Cells (Dry Cell) and Secondary Cells (Lead Acid battery).

Corrosion: Introduction to corrosion, Types of corrosion, Cause of corrosion, Corrosion prevention and control, Corrosion issues in specific industries (Power generation, Chemical processing industry, Oil & gas industry and Pulp & paper industries).

Chemistry of Engineering Materials:

Cement: Constituents, manufacturing, hardening and setting, deterioration of cement, Plaster of Paris (POP).

Electrochemistry: It is a branch of chemistry which deals with the transformation of chemical energy to electrical energy and vice versa.

Cell: A device which converts chemical energy to electrical energy and vice versa is known as cell.

Cells are of two types:

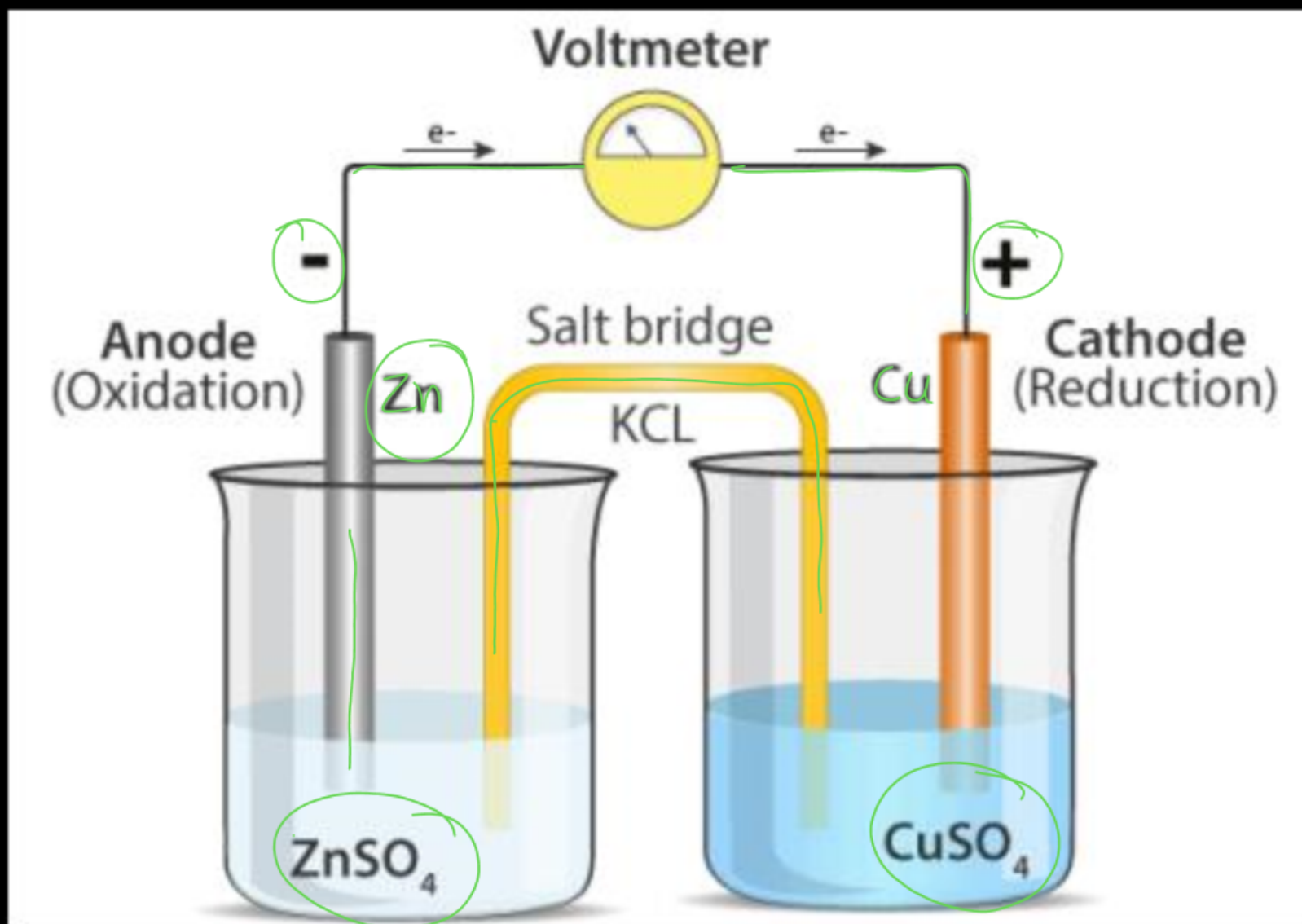
- (1) Electrochemical cell
 - (2) Electrolytic cell
- $\left\{ \begin{array}{l} \text{Anode} \rightarrow \text{Oxidation (loss of } e^-) \\ \text{Cathode} \rightarrow \text{Reduction (Gain of } e^-) \end{array} \right.$

Electrode potential: It is the measure of the tendency of metallic electrode to lose or gain electrons, when it is in contact with its own salt solution.

- (1) **Oxidation potential:** It is the measure of tendency of an electrode to lose electrons or to get oxidized is called its oxidation potential.
- (2) **Reduction potential:** It is the measure of tendency of an electrode to gain electrons or to get reduced is called its reduction potential.

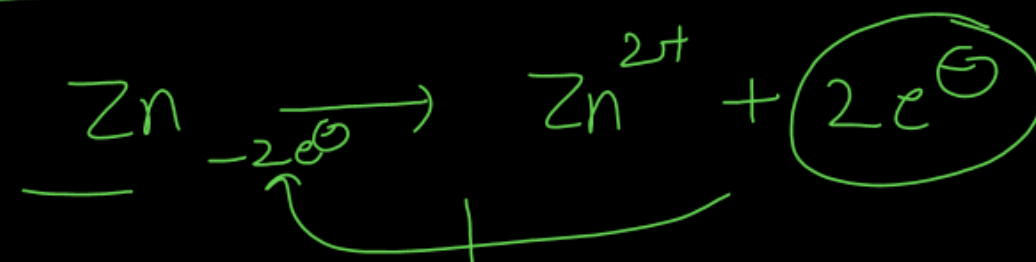
<u>Electrochemical cell or Voltaic Cell</u>	<u>Electrolytic cell</u>
✓ Cell which converts <u>chemical energy</u> to <u>electrical energy</u> .	Cell which converts <u>electrical energy</u> to <u>chemical energy</u> .
<u>Redox reaction</u> takes place which is <u>spontaneous</u> and it is responsible for the <u>production of electrical energy</u> .	<u>Redox reaction</u> is non spontaneous and <u>electrical energy</u> is <u>supplied</u> so that the reaction may take place.
✓ <u>Electrodes used are of dissimilar material</u> . <i>metal Zn Cu</i>	<u>Electrodes used may be of similar or different material</u> .
Each electrode is dipped in electrolytic solution of its own ions. <i>ZnSO₄ , CuSO₄</i>	<u>Both electrodes are dipped in same electrolytic solution</u> . <i>H₂SO₄</i>
<u>Salt bridge is used</u> .	<u>Salt bridge is not used</u> . X
✓ <u>Anode is negative</u> and <u>cathode is positive</u> .	✓ <u>Anode is positive</u> and <u>cathode is negative</u> .
✓ <u>Electrons move from anode to cathode</u> in external circuit	<u>Electrons</u> are supplied by battery and enter through <u>cathode</u> and come out through <u>anode</u> .
<u>Ex: Galvanic cell</u> .	Ex: Lead acid storage battery.

Electrochemical cell or Voltaic Cell



working

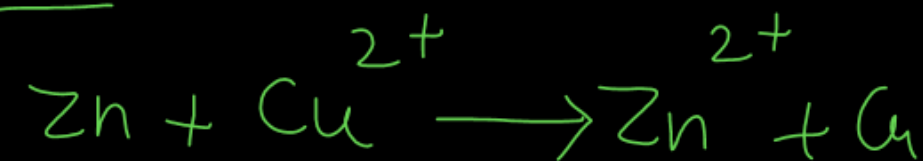
At anode :- (Oxidation)



At cathode :- (Reduction)



Overall Rxn



ELECTROMOTIVE FORCE OR CELL POTENTIAL (EMF)

EMF is the potential difference between the two electrodes of the cell when either no or negligible current is allowed to flow in the circuit.

$$\underline{E^0_{\text{cell}} = E^0_{\text{cathode}} - E^0_{\text{anode}}}$$

NERNST EQUATION

$$\Delta G = -n E_{\text{cell}} F \quad - (1)$$

$$\Delta G^\circ = -n E_{\text{cell}}^\circ F \quad - (2)$$

$$\Delta G = \Delta G^\circ + RT \ln Q \quad - (3)$$

Put values of ΔG and ΔG° in eqⁿ (3)

$$-n E_{\text{cell}} F = -n E_{\text{cell}}^\circ F + RT \ln Q \quad - (4)$$

Divide eqⁿ (4) by $-nF$

$$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{RT}{nF} \ln Q$$

$$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{RT}{nF} \times 2.303 \log_{10} Q$$

$$\ln Q = 2.303 \log_{10} Q$$

$$\left[\begin{array}{l} R \rightarrow \text{Gas Const.} = 8.314 \\ T \rightarrow \text{Temp} = 298\text{K} \\ F = \text{Faradays Const.} \\ 96500 \end{array} \right]$$

$$Q = \frac{\text{Conc of Products}}{\text{Conc of Reactants}}$$

Ques. Consider a cell reaction: $\text{Zn} / \text{Zn}^{2+}[0.1\text{M}] // \text{Cu}^{2+}[0.2\text{M}] / \text{Cu}$. Standard reduction potential of Zn^{2+} and Cu^{2+} are -0.76V and 0.34V respectively. Write half-cell reactions, complete cell reaction and calculate EMF of the cell. [2020-2021]

Solution



$$E^\circ_{\text{anode}} = -0.76\text{V}$$

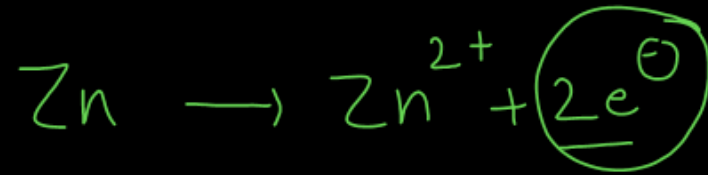


$$E^\circ_{\text{cathode}} = 0.34\text{V}$$

$$\log \frac{A}{B} = \log A - \log B$$

$\log 1 = 0$
 $\log 2 = 0.3010$

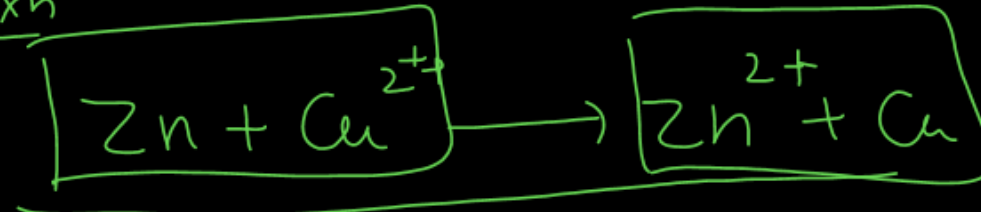
At anode:-



At cathode:-



Overall Rxn



$$Q = \frac{[\text{Zn}^{2+}][\text{Cu}]}{[\text{Zn}][\text{Cu}^{2+}]}$$

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{n} \log \frac{[0.1]}{[0.2]}$$

$$= 1.1 - \frac{0.0591}{2} (\log 1 - \log 2)$$

$$= 1.1 - 0.0296(0 - 0.3010) = 1.1 + 0.0296(0.3010)$$

$$E^\circ_{\text{cell}} = 0.34 - (-0.76)$$

$$= 1.10\text{V}$$

LEAD ACID STORAGE BATTERY

Construction:

- **Anode:** Made of lead grid filled with spongy lead.
- **Cathode:** Made of a lead grid filled with PbO_2 .
- **Electrolyte:** 28-30% solution of H_2SO_4 .

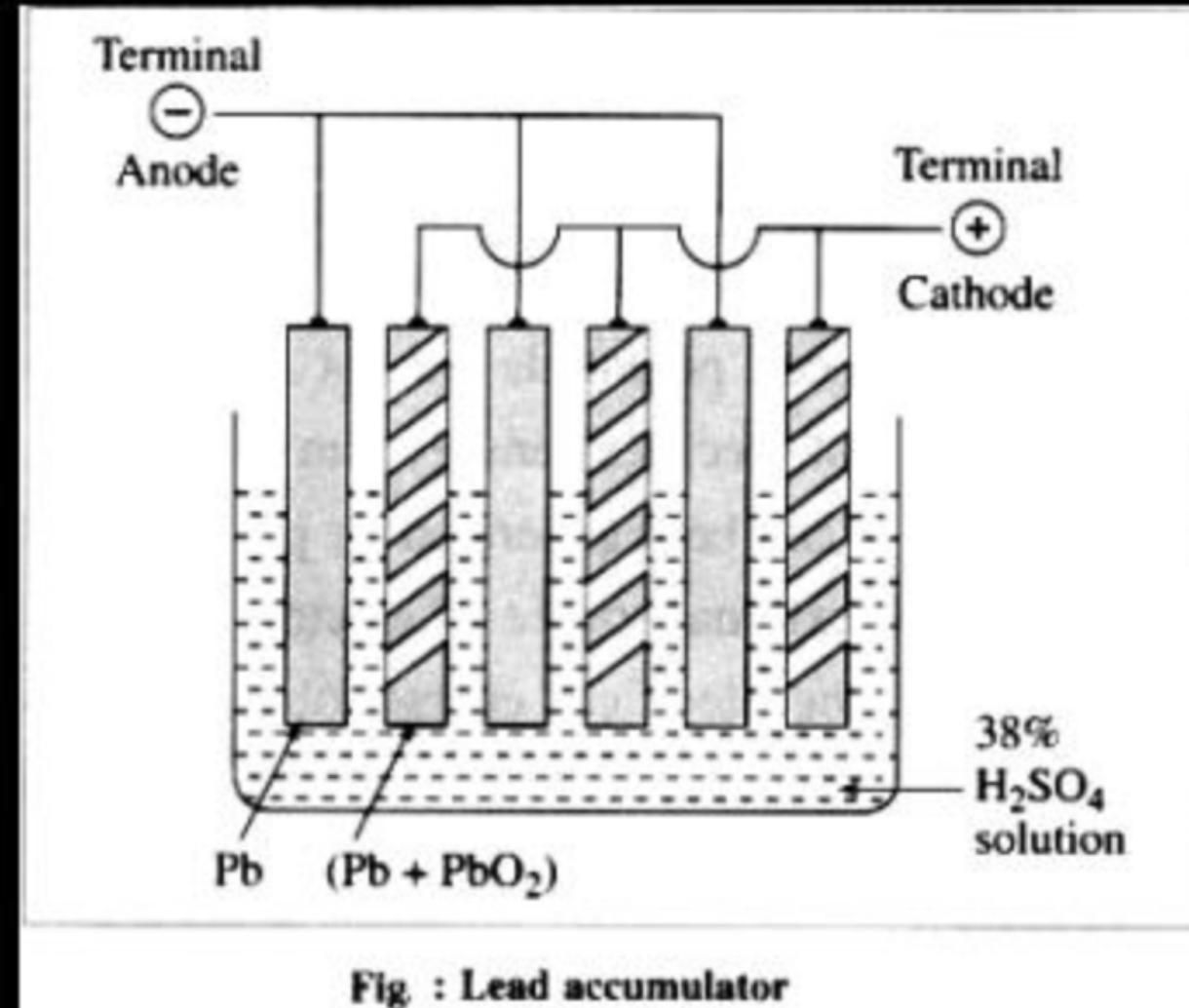


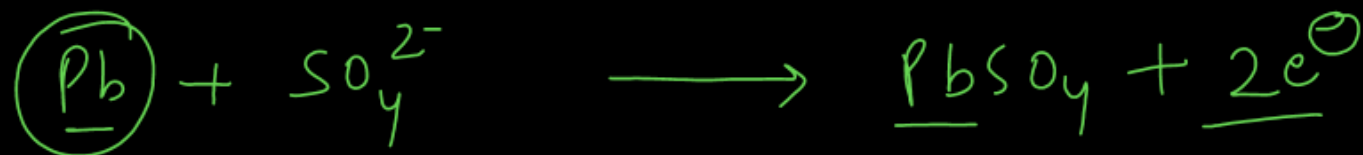
Fig : Lead accumulator

Working :-

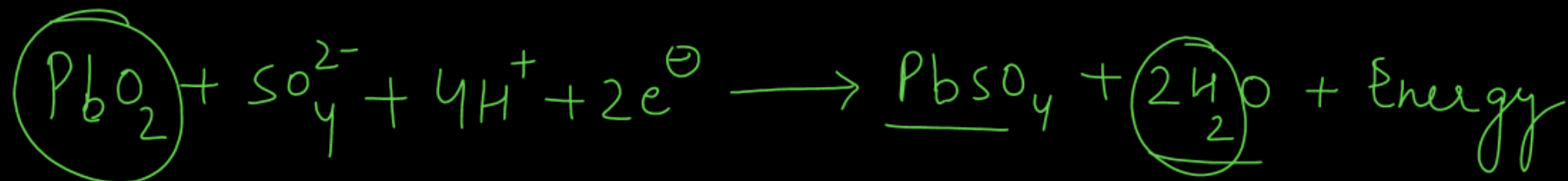
During Discharging :-

Electrochemical
cell

At anode :-
(-)



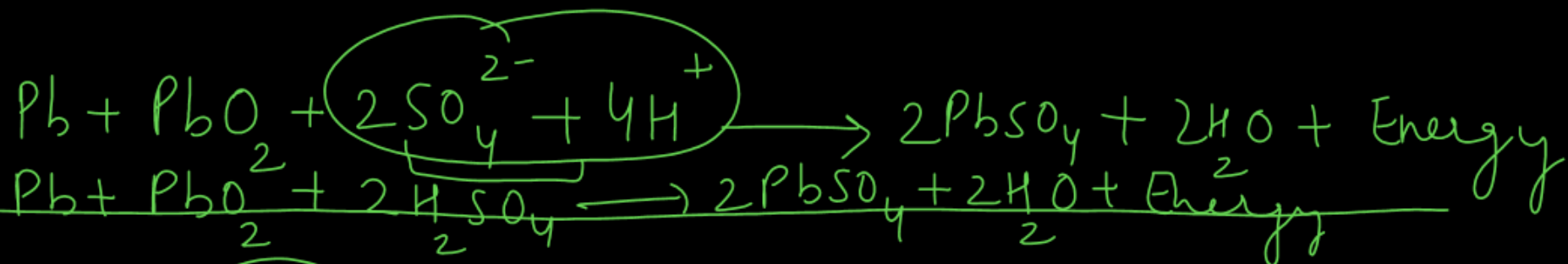
At cathode :-
(+)



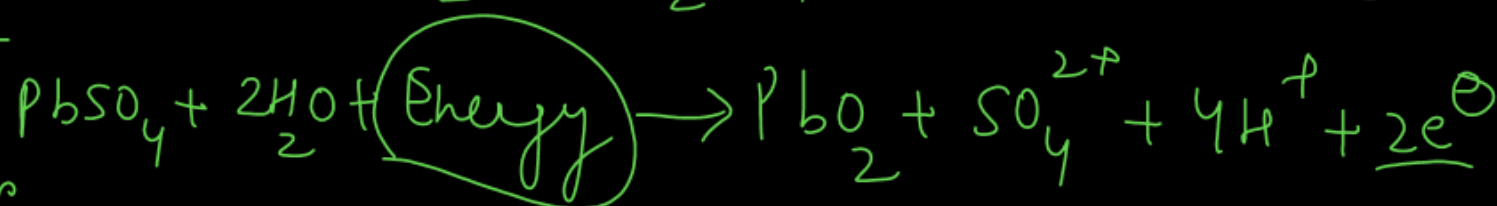
Overall Rn

During charging

OR



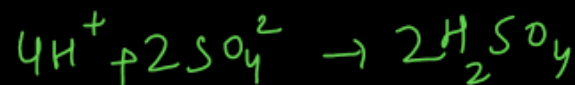
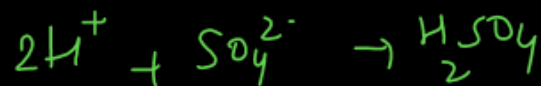
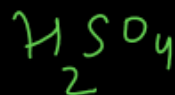
At anode :-



At Cathode :-



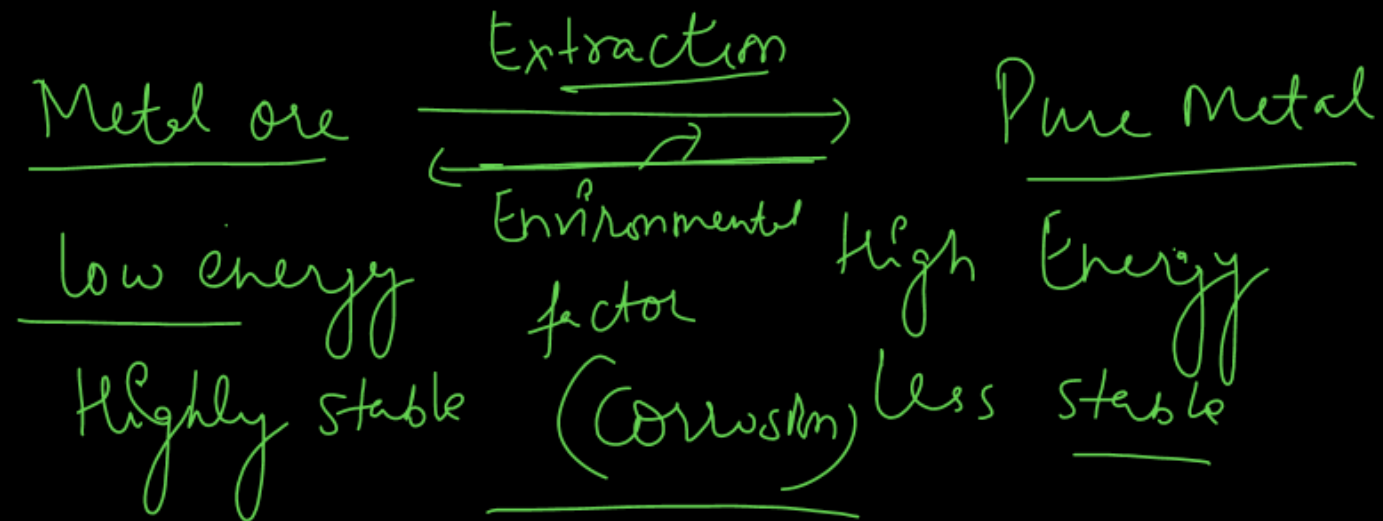
Overall Rn



Corrosion is a process of gradual destruction or deterioration of metals by chemical or electrochemical attack of environment.

Rusting of iron
Tarnishing of Silver

REASON / CAUSES OF CORROSION



FACTORS AFFECTING CORROSION:

- (1) Nature of metal
- (2) Nature of environment

(1) Nature of metal

(a) Position of metals in galvanic series

Highly reactive metals $\rightarrow$ Anode $\rightarrow$ Corrosion
less reactive metals $\rightarrow$ Cathode

(b) Areas of anode and cathode



(c) Purity of metal

Impurity leads to the corrosion

Active End



Passive End

Magnesium
 Magnesium alloys
 Zinc
 Aluminium alloys
 Cadmium
 Mild Steel
 Wrought Iron
 Cast Iron
 Stainless Steel (Active)
 Lead
 Tin
 Manganese
 Nickel (Active)
 Yellow Brass
 Red Brass
 Copper
 Nickel (Passive)
 Stainless Steel (Passive)
 Silver
 Graphite
 Gold
 Platinum

Anode

Cathode

(2) Nature of environment

(a) Temperature $\propto$ Corrosion

(b) Humidity $\propto$ Corrosion

(c) Moisture $\propto$ Corrosion

(d) pH acidic $\rightarrow$ corrosion
 Basic $\downarrow$

(e) Presence of impurities and suspended particles in atmosphere

Gaseous fumes, acid, NaCl, Na₂SO₄

TYPES OF CORROSION

(a) Dry Corrosion

(b) Wet Corrosion

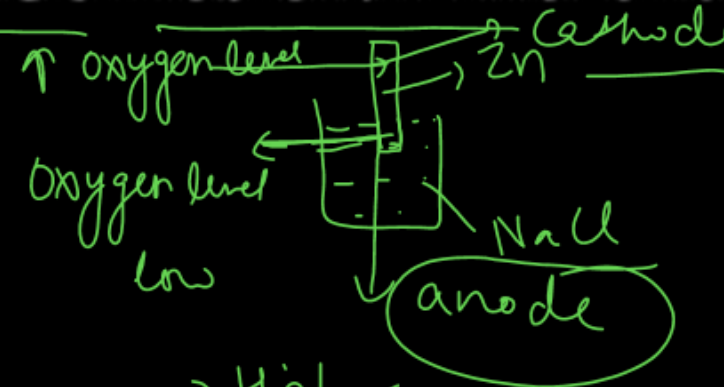
	Dry corrosion	Wet corrosion
1	It occurs in <u>dry condition</u> .	It occurs in <u>wet condition</u> .
2	If the corrosion takes place due to direct <u>chemical attack</u> (in the <u>absence of moisture</u>), <u>corrosion is known as dry corrosion</u> .	If the <u>corrosion</u> takes place due to <u>electrochemical attack</u> in <u>presence of moisture</u> or a <u>conducting medium</u> , <u>corrosion is known as wet corrosion</u> .
3	<u>Explained by absorption mechanism</u>	<u>Explained by electrochemical mechanism</u>
4	It occurs on both <u>heterogeneous</u> and <u>homogeneous</u> surfaces.	It occurs only on <u>heterogeneous metal surfaces</u> .
5	Corrosion is <u>uniform</u> .	Corrosion is not uniform. X
6	It is a <u>slow process</u> .	<u>It is a fast process</u> .
7	<u>Corrosion products accumulate at the place where corrosion occurs</u> .	<u>Corrosion take place at anode</u> but products <u>accumulate near the cathode</u> .

CLASSIFICATION OF CORROSION

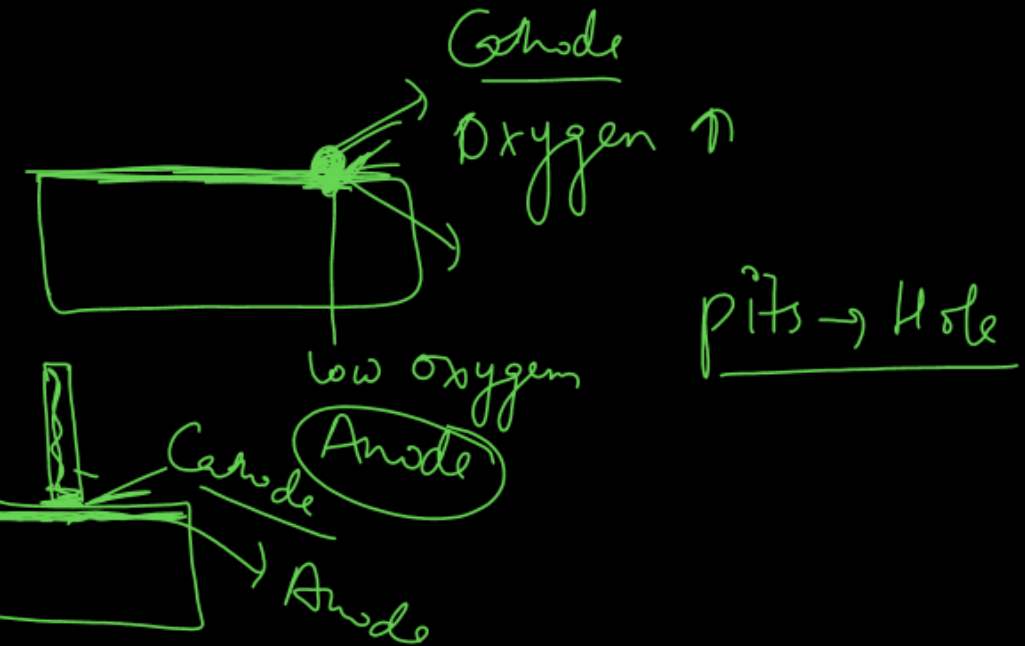
(a) Galvanic Corrosion/Bimetallic Corrosion/Differential metal Corrosion



(b) Concentration Cell Corrosion/Differential Aeration Corrosion



(c) Pitting Corrosion



(d) Crevice Corrosion



(e) Stress Corrosion

WET THEORY OF CORROSION

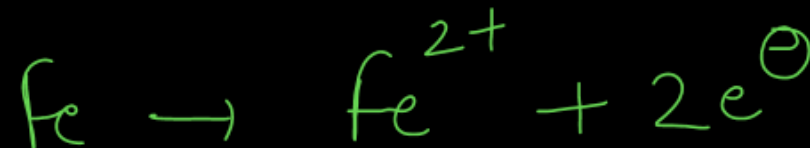
(Rusting of Fe)

(a) Oxygen Absorption Mechanism (In basic or neutral medium)

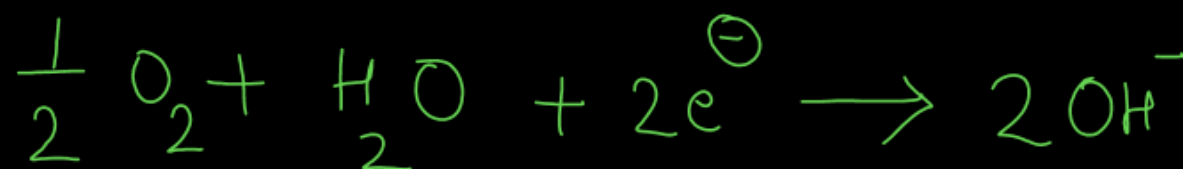
(b) Evolution of Hydrogen Mechanism (In Acidic Medium)

Oxygen absorption Mechanism

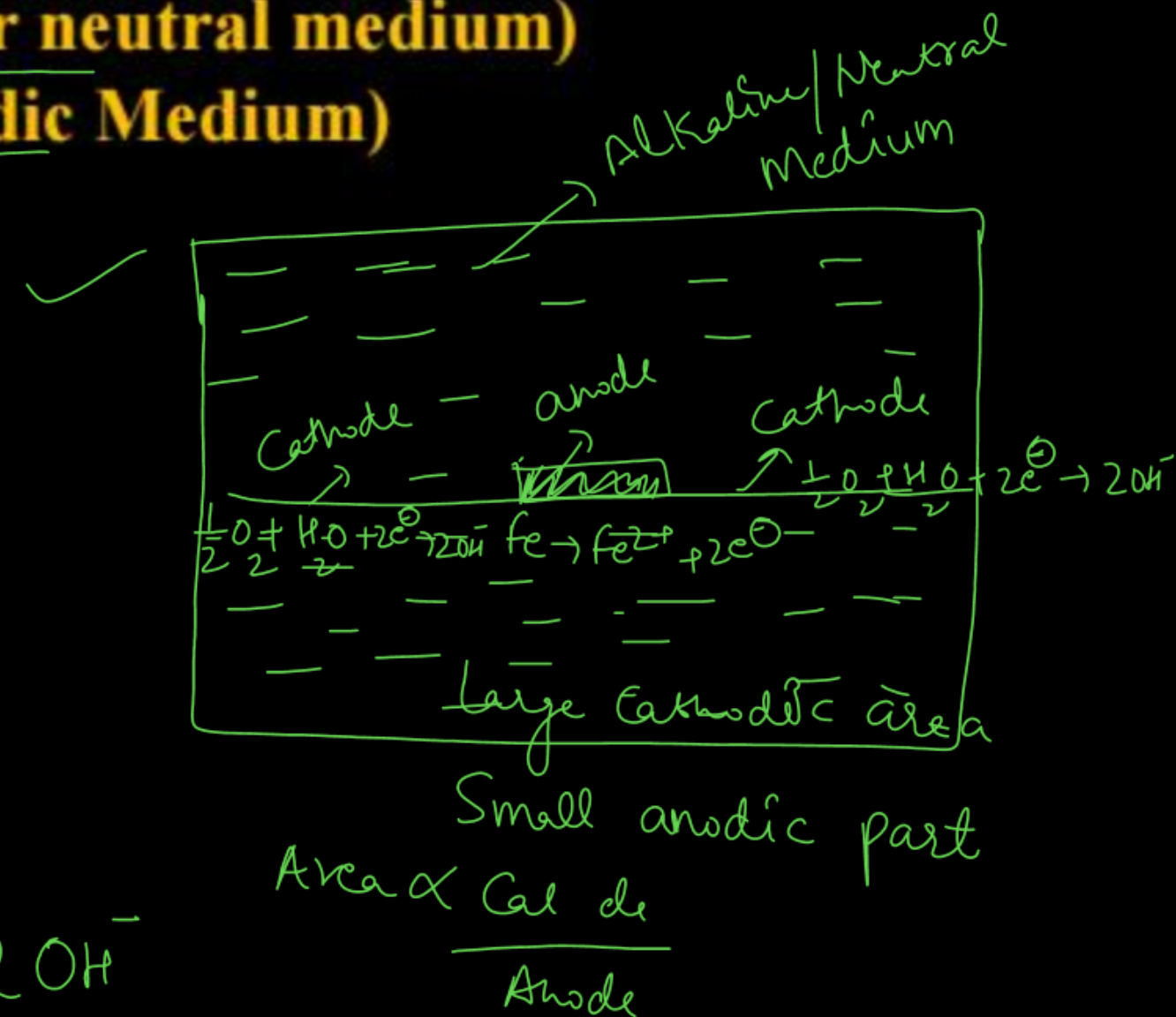
At anode :-



At Cathode :-



Overall Rn

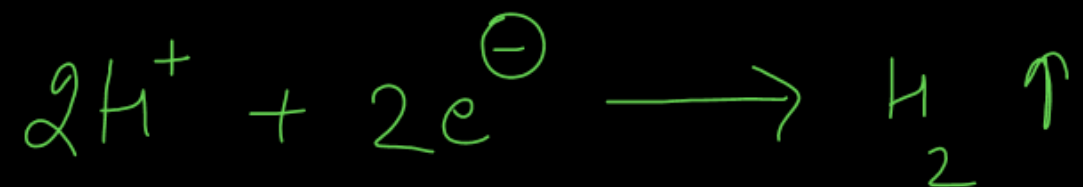


(2) Hydrogen evolution (Acidic medium) H^+

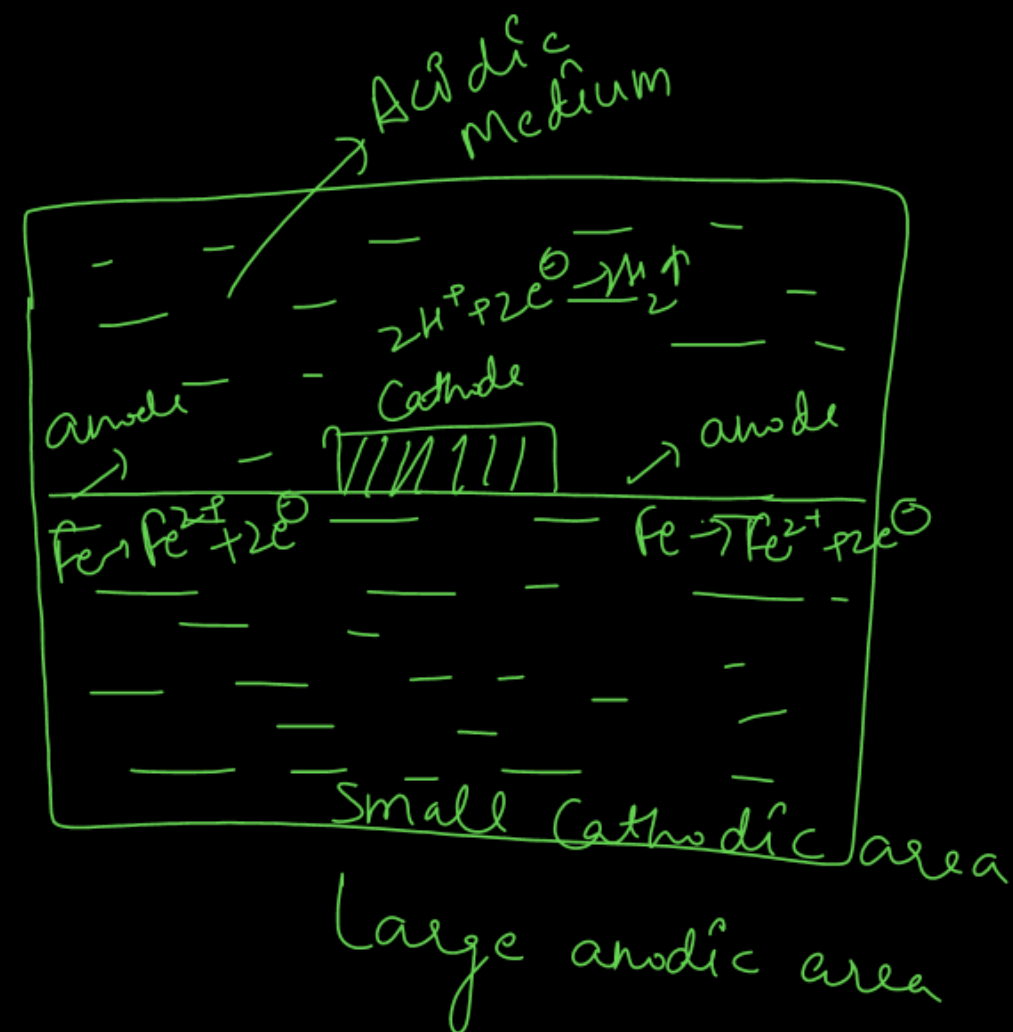
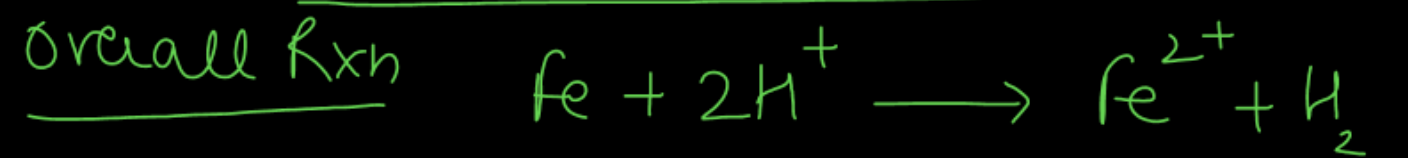
At anode :-



At Cathode :-



Overall Rxn



PREVENTION OF CORROSION

(1) PROPER DESIGN

- a) Avoid sharp bends and sharp corners.
- b) Avoid contact of dissimilar metals in presence of corroding medium.
- c) There should be proper drainage.
- d) When the two dissimilar metals are to be used in contact, anodic area should be large and cathodic area should be small.
- e) If two dissimilar metals have to be used in contact, their position in electrochemical series should be close as possible.
- f) If two metals have to be coupled use insulation between them.



$\text{Area} \propto \frac{\text{Cathode}}{\text{Anode}}$

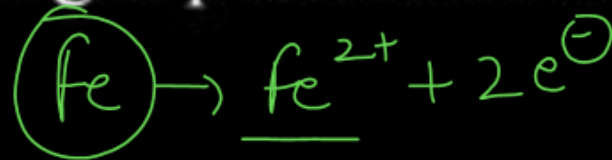
Highly reactive
↓
less reactive

(2) BY USING INHIBITORS

The substances which are added from outside to inhibit the corrosion are known as inhibitors.

a) ANODIC INHIBITORS

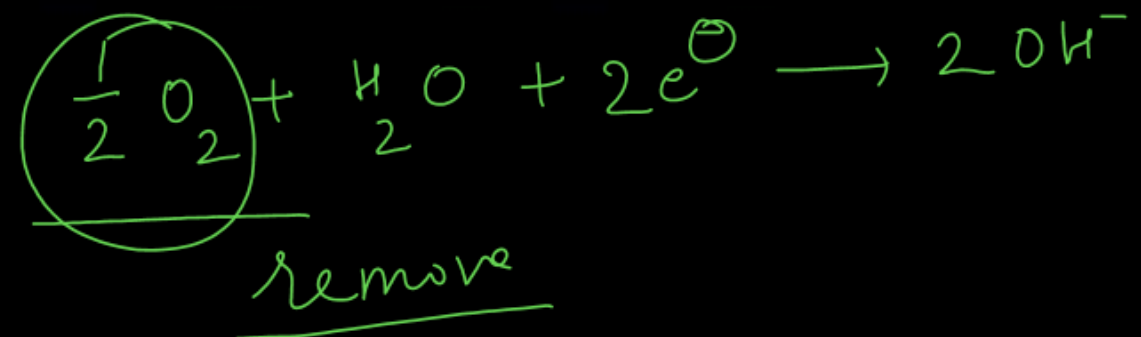
They react with the metallic ions of anode and form insoluble precipitate which is adsorbed on metal surface forming a protective film/barrier. Ex: alkalis, phosphates, chromates, etc.



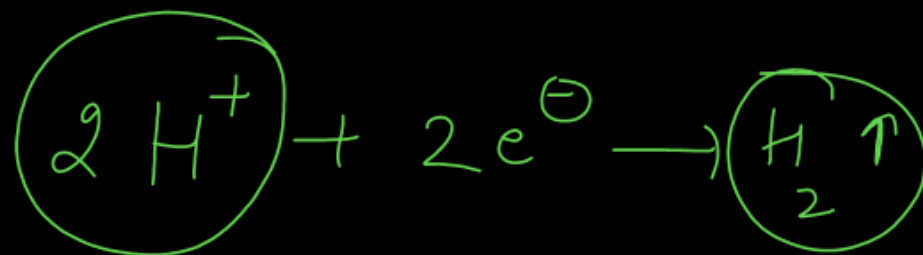
b) CATHODIC INHIBITORS

They decrease the rate of reaction thereby reducing rate of overall corrosion.

i) In Neutral/Alkaline Medium



ii) In Acidic Medium



H⁺ diffusion

PREVENTION OF CORROSION

Corrosion can be prevented by various protective measure such as:

- (1) Modification of the Environment.
- (2) Modification of the Properties of Metal.
- (3) Use of surface coating/Protective Coating
- (4) Cathodic Protection
- (5) By proper selection of Material and Design

(1) Modification of the Environment

(a) Removal of Corrosion Stimulants

- **✓ Acid**
- **✓ Salts**
- **✓ Moisture**
- **✓ Oxygen**

(b) Corrosion Inhibitors

- **✓ Anodic Inhibitors**
- **✓ Cathodic Inhibitors**
- **✓ Vapour Phase Inhibitors/Volatile Inhibitors**

(2) Modification of the Properties of the Metal

- Alloying
- Annealing
- Refining

(3) Use of the Protective Coating

(a) Metallic Coating

- Anodic Coating
- Cathodic Coating

(b) Organic Coating

(c) Inorganic Coating

Base metal prevent → Metal coating

Anode

Cathode

✓ Anodic Coating	✓ Cathodic Coating
✓ Protects the metal from corrosion <u>sacrificially</u> .	Protects the metal from corrosion due to higher <u>corrosion</u> resistance. ✓
✓ Base metal <u>is not corroded</u> , even if it is <u>exposed</u> till all the coating gets consumed.	Corrosion of base metal is speeded up of <u>coating breaks</u> and metal gets <u>exposed</u> .
✓ Example: <u>Coating of Zinc</u> on <u>Iron</u> .	Example: <u>Coating of Tin</u> on <u>Iron</u> .

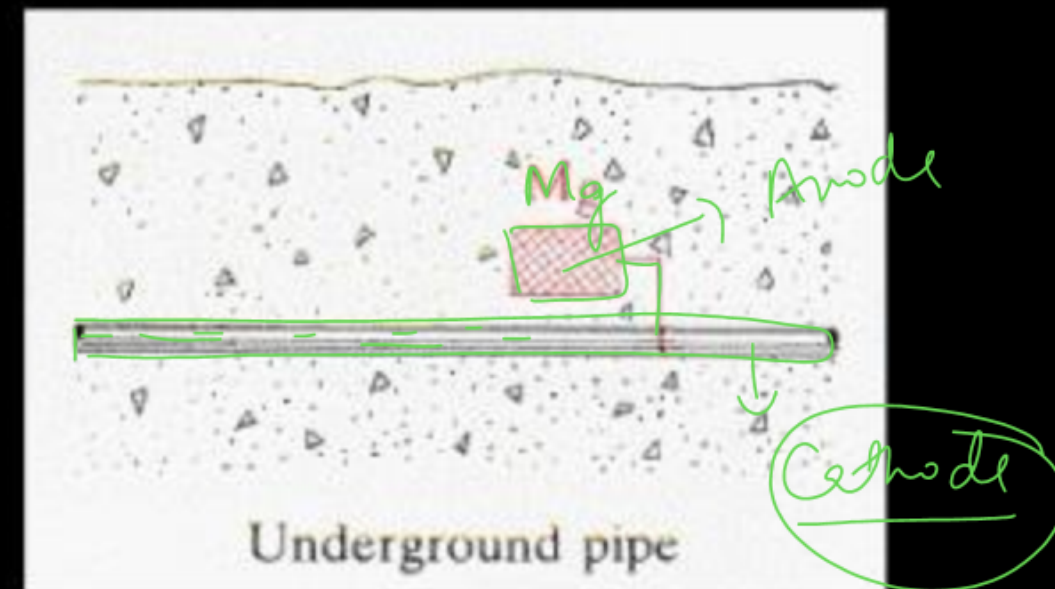
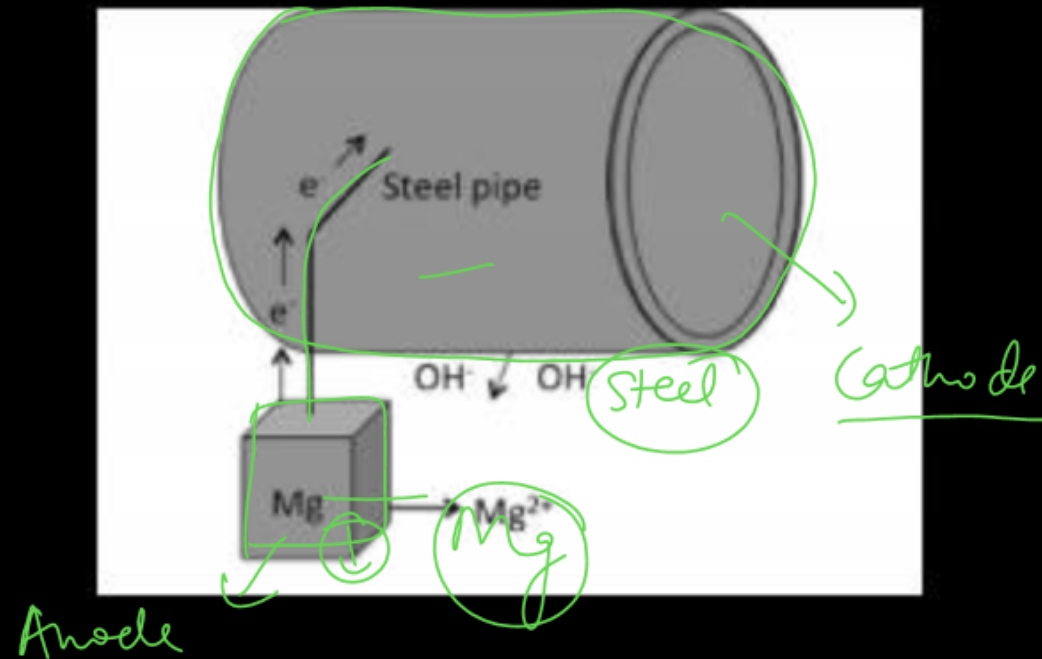
(4) Cathodic Protection

The method of protecting the base metal by making it to behave like a cathode is called as cathodic protection.

- Sacrificial Anodic Protection
- Impressed Current Cathodic Protection

SACRIFICIAL ANODIC PROTECTION

Base metal is connected by a wire to a more reactive metal so that the base metal becomes cathode and more reactive metal becomes anode.

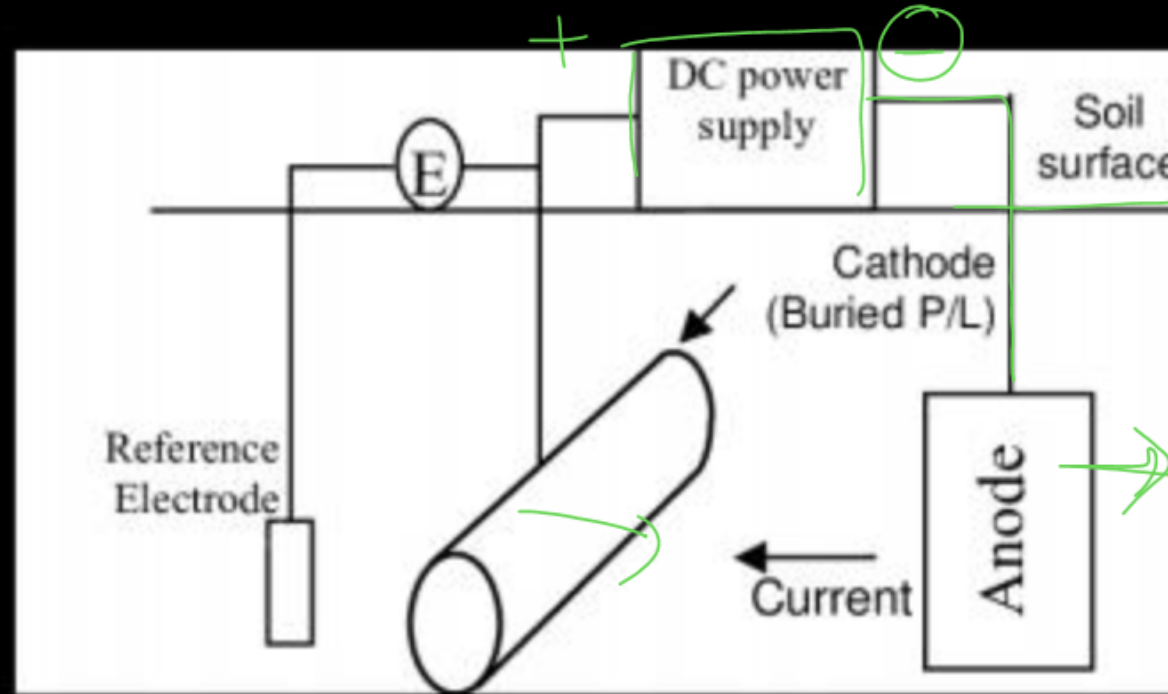


Ex: The underground water pipelines and water tanks are also protected by sacrificial anode method.

IMPRESSED CURRENT

In this method, an impressed current is applied in opposite direction to nullify the corrosion current, and convert the corroding metal from anode to cathode.

The impressed current is slightly higher than the corrosion current. Thus the anodic corroding metal becomes cathodic and protected from corrosion.



The impressed current protection method is used for water tanks, water & oil pipe lines etc.

(5) By proper selection of Material and Design

Galvanneal Steel Metal

CEMENT

Constituents of Cement:

1. Calcium Oxide or lime (CaO)	50-60%
2. Silica (SiO_2)	20-25%
3. Alumina (Al_2O_3)	5-10%
4. Magnesia (MgO)	2-3%
5. Ferric Oxide (Fe_2O_3)	1-2%

Manufacture of Portland Cement:

The steps involve in the manufacturing process are as follows:

- ✓ 1) Mixing
- ✓ 2) Burning
- ✓ 3) Grinding
- ✓ 4) Packing

(1) Mixing:

Mixing of raw materials can be done either by (a) dry process (b) wet process

<u>Dry Process</u>	<u>Wet Process</u>
✓ It is adopted when the raw materials are quite hard.	It can be used for any type of raw materials.
✓ Fuel consumption is low.	Fuel consumption is high.
✓ Process is slow.	✓ Process is comparatively faster.
✓ Cement produced is of inferior quality.	✓ Cement produced is of superior quality.
Cost of production of cement is less.	Cost of production of cement is somewhat higher.

(2) Burning:

Rotary Kiln

Upper part

Medium

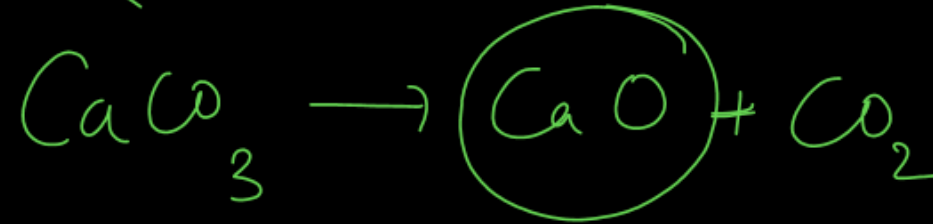
Lower

(a) Drying zone:

400°C

(b) Calcinations zone:

$(1000^{\circ}\text{C} = 1200^{\circ}\text{C})$



(c) Clinkering zone:

lower part of Rotary kiln
(1200°C to 1700°C)

(3) Grinding

Grind



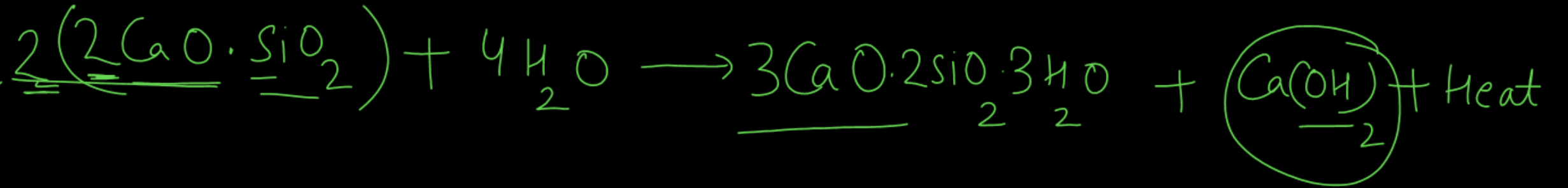
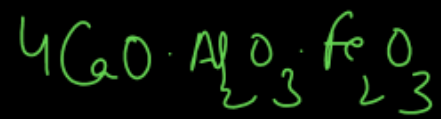
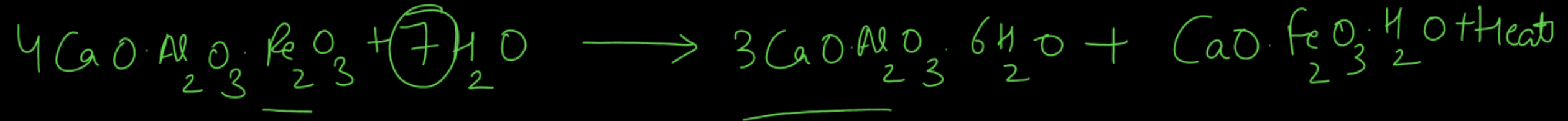
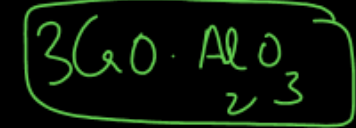
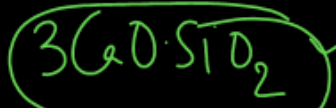
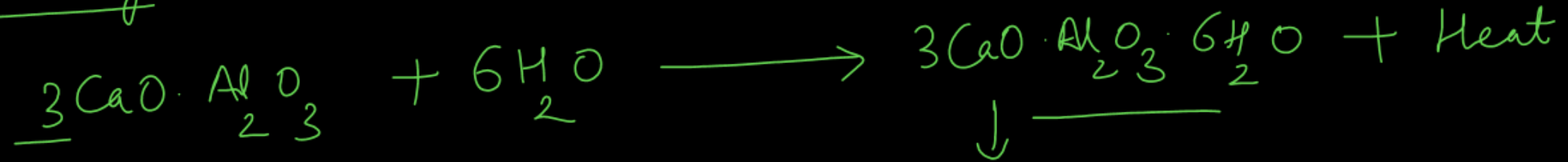
(4) Packing

2-3% Gypsum

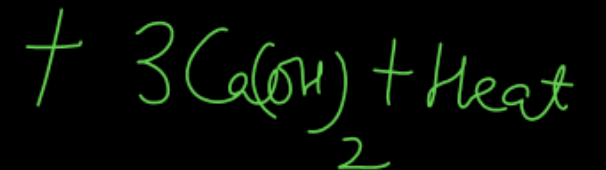
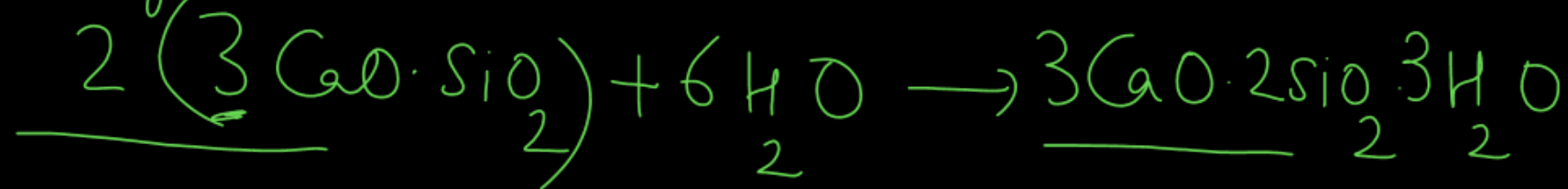
SETTING AND HARDENING OF CEMENT

Initial setting

③ ④ ① ②
1 1 1 1



Final setting



Download **Gateway Classes** Application
From Google Play store

Link in video Description

Thank You